

HW #2

59.6%

1. a)

b)

c)

d)

2. a) $10.02 \text{ bar} = 1.002 \text{ MPa}$

b) $x = \frac{V - V_f}{V_{fg}} \quad V_f = 1.1274 \quad V_g = 0.1941 \quad V_{fg} = V_g - V_f = -0.933$

$x = \frac{V - 1.1274}{-0.933}$

3. a) $\frac{T_1 - T_2}{T_1 - T_x} = \frac{P_1 - P_2}{P_1 - P_x} \rightarrow \frac{130 - 140}{130 - T_x} = \frac{2.701 - 3.613}{2.701 - 3} \rightarrow \frac{-10}{130 - T_x} = 3.051$

$\rightarrow 130 - T_x = -3.279 \rightarrow T_x = 133.279^\circ \text{C}$

b) $v_2 = \frac{V_2}{m} = \frac{2.064 \text{ m}^3}{2 \text{ kg}} = 1.032 \text{ m}^3/\text{kg}$

$\frac{T_1 - T_2}{T_1 - T_x} = \frac{v_1 - v_2}{v_1 - v_x} \rightarrow \frac{110 - 120}{110 - T_x} = \frac{1.210 - 0.8919}{1.210 - 1.032} \rightarrow \frac{-10}{110 - T_x} = 1.787$

$110 - T_x = -5.596 \rightarrow T_x = 115.596^\circ \text{C}$

c) $V_1 = v_1 \times m = 0.6058 \times 2 = 1.2116$

$W = -P(V_2 - V_1) = -3(2.064 - 1.2116) = -2.56 \text{ kJ}$

4. a) $v_1 = 0.02301 \text{ m}^3/\text{kg}$

$V_1 = v \times m = 0.02301 \times 1 \text{ kg} = 0.02301 \text{ m}^3$

$u = 268.35 \text{ kJ/kg}$

$U = u \cdot m = 268.35 \text{ kJ}$

$U = Q + W \rightarrow 268.35 = Q + 20 \rightarrow Q = 248.35 \text{ kJ}$

5. a) $P_1 = 0.3119 \text{ bar} = \boxed{31.19 \text{ kPa}}$ ✓

b) $u_g = 2469.6 \text{ kJ/kg} \rightarrow U = u_g \times m = 2469.6 \times 1.5 = 3704.4 \text{ kJ}$

$W = -P(V_2 - V_1) = 0 \rightarrow \boxed{U = Q = 3704.4 \text{ kJ}}$ ✓

6. a) $V_f = v_f \times m = (0.8352 \times 10^{-3}) \times 25 = 0.02088 \text{ m}^3$

$V_g = v_g \times m = 0.0236 \times 25 = 0.59 \text{ m}^3$

$V = V_f + V_g = 0.61088 \text{ m}^3$

$v = 0.8352 \times 10^{-3} + 0.6(0.0236 - 0.8352 \times 10^{-3}) = 0.0145$

$\boxed{V = v \times m = 0.3625}$

b) $V_{\text{vapor}} =$

~~0.59~~

~~0.59~~

7. a) $R = \frac{\bar{R}}{M} = \frac{8.3145 \text{ kJ/kmol} \cdot \text{K}}{4.003 \text{ kg/kmol}} = 2.077 \text{ kJ/kg} \cdot \text{K} = \boxed{2.077 \times 10^{-3} \text{ J/g} \cdot \text{K}}$

c) $PV = mRT \rightarrow (0.8 \times 10^5 \text{ Pa})V = (0.2 \text{ g})(2.077 \times 10^{-3} \frac{\text{J}}{\text{g} \cdot \text{K}})(5.2 \text{ K})$
 $\rightarrow (0.8 \times 10^5 \text{ Pa})V = 0.00216 \text{ J} \rightarrow V = 2.7 \times 10^{-8} \text{ m}^3 = \boxed{0.027 \text{ cm}^3}$

b) $PV = mRT \rightarrow (1.3 \times 10^5) V = (0.2)(2.077 \times 10^{-3})(5.2)$
 $\rightarrow V = 1.66 \times 10^{-8} = \boxed{0.0166 \text{ cm}^3}$

~~0.0166~~

8. a) $R = \bar{R}/M = 8.3145/44.01 = 0.189$

$PV = mRT \rightarrow (2.75 \text{ kPa})(0.05 \text{ m}^3) = m(0.189 \frac{\text{kJ}}{\text{kg} \cdot \text{K}})(298.15 \text{ K})$

$\rightarrow \boxed{m = 0.244 \text{ kg}}$ ✓

b)

~~0.244~~